

THE TRANSMISSION OF RELIABLE AND UNRELIABLE INFORMATION*

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Abstract

Information often spreads and influences beliefs regardless of its reliability. We show that this occurs in part because indicators of reliability tend to be lost in the process of word-of-mouth transmission. We conduct controlled experiments where participants listen to economic forecasts and pass them on through voice messages. Other participants listen either to original or transmitted audio recordings and report incentivized beliefs. Across various transmitter incentive schemes, a claim's reliability is lost in transmission much more than the claim itself. Reliable and unreliable information, once filtered through transmission, impact listener beliefs similarly. Mechanism experiments show that reliability is lost not because it is perceived as less relevant or harder to transmit, but because it is less likely to *come to mind* during transmission. Evidence from experiments, a large corpus of everyday conversations, and economic TV news demonstrates that reliability information often fails to be conveyed because it is less likely to be cued during transmission.

Keywords: Information Transmission, Word-of-mouth, Reliability, Memory.

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1 Introduction

The extent to which information spreads and influences beliefs is often unrelated to its reliability. Consider three examples. The “4% rule” for retirement savings, a standard piece of financial advice, originated as a descriptive finding in a 1994 analysis of historical data that the author warned were unrepresentative and dangerous to extrapolate from. The author later expressed surprise at how quickly his caveats dropped away as the finding passed from mouth to mouth and solidified into a “rule” (Bengen, 2024). One of the most viral health tips of the 2000s claimed that eating six small meals a day promoted weight loss. The tip was repeated on broadcast television shows, in bestselling books, and throughout the multi-billion-dollar weight loss industry (ABC News, 2005; Gower, 2002). Those repeating the claim rarely said anything about its provenance or evidentiary basis; it appears to have originated either in a 1967 book by a physician who claimed sensationally to have helped 10,000 patients lose weight, or in a 1991 randomized controlled trial with a sample size of 7 (Stillman and Baker, 1967; Tai et al., 1991). A 2014 RAND retrospective about the 2002 National Intelligence Estimate on weapons of mass destruction in Iraq concluded that the report “contained several qualifiers that were dropped” when it was transformed into an executive summary, and that “as the draft NIE went up the intelligence chain of command, the conclusions were treated increasingly definitively” (Gompert et al., 2014).

Economic theory predicts that more reliable information should influence beliefs more. The preceding anecdotes, however, suggest that unreliable claims frequently influence behavior too much, and reliable claims too little, in part because the claims themselves get passed from person to person while information about their credibility or evidentiary basis is lost in transmission.

This paper studies whether, and why, this is true. Borrowing standard terminology, we distinguish between the realization of a signal (the “level” of a claim) and the precision of that signal (the “reliability” of the claim). This distinction gives us a natural benchmark: we ask whether the precision of signals (reliability) tends to be lost in word-of-mouth transmission *more* than the signal realizations (level). We then ask whether the information people transmit is influenced by cost-benefit calculations about the payoffs to transmitting different kinds of information, or by cognitive constraints such as selective attention or memory. To study these questions, we use controlled online experiments and observational data from everyday conversations and TV news.

Design. Our experiments involve more than 9,000 participants. In a *transmitter* experiment, participants listen to a one-minute message giving a qualitative forecast about an economic variable, and are incentivized to record themselves faithfully passing on the information they heard. In a subsequent *listener* experiment, participants listen to either the original forecasts or transmitted versions of those forecasts before stating incentivized beliefs.

We base the original forecasts on actual news coverage, and cross-randomize their level and reliability. We vary the *level*—high or low—by switching whether the original message argues for an increase or a decrease in the relevant variable. We vary the *reliability*—reliable or unreliable—using manipulations that weave certainty- or uncertainty-denoting words into an otherwise-identical text, and manipulations that change multiple implicit and explicit signals

of reliability, including the speaker’s confidence, credentials, stated sources of evidence, fluency, and vocabulary. Both level and reliability are communicated in qualitative terms only, mimicking how people naturally communicate. Transmitters are incentivized based on how close the belief updates induced by their voice messages are to the average belief updates induced by the original messages. Such incentives motivate a faithful transmission of information, which is ubiquitous in the real world: sales employees relay customer feedback to developer teams, analysts brief executives, and friends share financial advice sourced from media.

Main results. Our main finding is that information about the reliability of a prediction is lost in transmission more than twice as much as information about the prediction’s level. We refer to this finding as *differential information loss*, and document it using three sets of analyses.

In our first analyses, we examine the transcripts of transmitted messages. Nearly all of the messages include some statement about the level of the original prediction, but only a third mention the original prediction’s reliability or include other markers of reliability, such as uncertainty prefixes. Transmitted messages, containing an average of 114 words (8-10 sentences), tend to be only half as long as the original messages. Yet even the longest decile of transmitted messages, which are about as long as the original messages, mention reliability less than 30% of the time. Many messages go on at length, and in great detail, about the level of the original forecast without mentioning its reliability.

What ultimately matters is the preservation of the information content of the original messages, which our transcript analysis does not fully capture: the original information could be preserved using fewer or different words. We therefore analyze two sets of beliefs: listeners’ beliefs about the level and reliability of the original messages (“message beliefs”) and their belief updates about the underlying economic state discussed in the recordings (“state belief updates”).

We begin with message beliefs. Among listeners who directly hear the original messages, a shift from a low-level to a high-level message changes beliefs about the prediction’s level by 1.37 standard deviations (SDs). Among listeners who hear transmitted versions of those messages, beliefs shift by only 0.88 SDs. This indicates a $100 \times [(1.37 - 0.88) / 1.37] \approx 34\%$ loss of sensitivity to variation in the level of the original prediction. By contrast, loss of reliability information is nearly three times as large. Among listeners who hear the original messages, switching from a weak-reliability to a strong-reliability message shifts beliefs about the message’s reliability by 1.18 SDs. The corresponding shift for listeners who hear transmitted recordings is 0.12 SDs, meaning 90% of the variation in information about a message’s reliability is lost in transmission.

In both cases, the loss of sensitivity to our manipulations is driven by a *symmetric* compression of beliefs towards an intermediate value. After transmission, high forecasts are perceived as predicting a less high level, and low forecasts as predicting a less low level. Similarly, reliable messages are perceived as less reliable, but unreliable messages are perceived as more reliable.

We then turn to state belief updates, that is, listeners’ belief updates about the economic variables discussed in the recordings. State belief updates are the object of ultimate economic relevance and the object on which transmitters are incentivized. Listeners who directly hear the original messages update their beliefs in a qualitatively Bayesian way: they update in the di-

rection of the message’s prediction, and those who hear strong-reliability versions of a message update twice as strongly, on average, as those who hear weak-reliability versions. By contrast, listeners who hear transmitted versions of the messages update about the same amount on average from weak-reliability and strong-reliability messages—the distinction between weak- and strong-reliability messages is almost completely lost in transmission. Level and reliability information loss interact: for highly reliable original messages, loss of level and reliability information push in the same direction, strongly attenuating belief updates towards zero. By contrast, for unreliable original messages, the two types of loss counteract each other, resulting in smaller effects on belief updates. This suggests that the primary consequence of the phenomenon we document is to drag down the influence of reliable information relative to unreliable information.

Mechanisms. We next ask what drives the differential information loss we document. On the one hand, reliability information could be lost more than level information as the result of a deliberate tradeoff, either because the perceived benefits of transmitting reliability information are lower or the perceived cognitive costs are higher. On the other hand, differential loss could result from some non-deliberate mechanism. For example, reliability information might not *come to mind* at the moment of recording the voice message. In a series of mechanism experiments, we reject the first two explanations and find support for the third.

We begin by studying beliefs about the benefits of passing on level versus reliability information. When asked, transmitters in our main experiment on average deem both equally important. We also conduct an additional experiment that explicitly and equally incentivizes transmitters to pass on level and reliability information, effectively holding beliefs about the relative benefits of transmitting the two dimensions constant. Even under these conservative incentives, we find pronounced differential information loss (34% for level and 70% for reliability).

Next, we ask whether the perceived cognitive costs of transmitting reliability information are higher. We conduct an experiment where transmitters are allowed to decide whether their bonus payment will depend on their transmission of level information or reliability information. *Ex ante*, a majority choose to be incentivized based on their transmission of *reliability* information and expect it to be easier to communicate, a view that does not change after having experienced the task, suggesting that higher perceived costs of transmitting reliability cannot explain our results.

Finally, we extend our analysis of mechanisms beyond perceived benefits and costs to examine the possibility that reliability information may simply fail to come to mind during the cognitively taxing task of information transmission. Our previous results hint at this possibility: more than 60% of transmitters do not mention reliability information at all in their messages, even when *ex post* remembering this information, saying that it is equally important as level information, and believing it is easier to transmit.

Drawing on workhorse frameworks of episodic memory (Bordalo et al., 2025b; Kahana, 2012), we focus on the possibility that reliability is less likely to come to mind because it is less likely to be *cued* in transmission contexts. Several pieces of evidence show that reliability is much more likely to be passed on when brought to the transmitter’s attention via contextual cues. This is true for a range of cue types. It is true for indirect, payoff-irrelevant cues of reliability

incidentally varied in our baseline experiments. It is true for naturally occurring cues in everyday language and on economic TV news coverage. Finally, it is true for explicit and highly salient cues in a supplementary experiment where we replicate our baseline design but confront transmitters with a prominent reminder to transmit reliability on the screen where they complete their recordings. In this final experiment, differential information loss is eliminated entirely. Despite the importance of cues, we show using a large dataset of spoken conversations that requests for information virtually never explicitly ask about its reliability, suggesting that generic requests for information transmission may not typically be associated with (and hence cue) reliability.

In summary, our mechanism analyses suggest that failures to pass on reliability information do not stem from transmitters representing the transmission problem correctly but consciously choosing to place lower weight on the transmission of reliability information; rather, they stem from transmitters having an incomplete representation of the transmission problem in the sense of Bordalo et al. (2025b), due to the typical absence of contextual factors cueing reliability.

Related literature. Our results show that reliable and unreliable information, once filtered through transmission, become hard to distinguish. This effect of transmission may operate alongside and compound a distinct *updating* bias: Griffin and Tversky (1992), Massey and Wu (2005), and Augenblick et al. (2025), among others, argue that even conditional on knowing the precision or diagnosticity of a signal, people overinfer from unreliable signals and underinfer from reliable ones.¹ In our experiments, the effect of such an updating bias is held constant across listeners to original and transmitted messages by design, allowing us to identify the distinct effect of transmission. Jointly, transmission-induced loss of reliability indicators and people’s insufficient sensitivity to received reliability indicators may reduce the influence of reliable and trustworthy information relative to unreliable information. The transmission loss we document also has a distinct consequence. While individual belief updating biases are privately costly, they do not necessarily affect others if they do not influence the way signal realizations are repeated to peers. By contrast, failures to pass on reliability information impose externalities, including on sophisticated peers who *would* respond appropriately to reliability information if it reached them. We demonstrate these consequences in simulation exercises calibrated to match results from our experiments, which we also use to explore the consequences of transmission-induced information loss for patterns of belief convergence and polarization.

Our focus on the transmission of qualitative stories about economic variables relates to a growing literature on the diffusion of narratives (Hirshleifer, 2020; Shiller, 2017). The loss of reliability we document in oral transmission may be part of a broader hypothesized phenomenon (e.g., Hirshleifer, 2020): as stories get told and re-told, they are simplified in the specific sense that nuance is lost. Recent contributions in this literature have focused on the role of narratives for belief formation (Andre et al., 2025; Barron and Fries, 2024; Graeber et al., 2024a,b; Kendall and Charles, 2025). Other papers that focus on the incentives governing the supply of information include Thaler (2025), Thaler et al. (2025), and Serra-Garcia (2025).

¹Note that signal “strength” in the model of Griffin and Tversky (1992) is related to level information in our framework, and signal “weight” to reliability information.

We also relate to a literature on how belief formation is shaped by selective attention (Ba et al., 2024; Graeber, 2023; Hartzmark et al., 2021), memory (Bordalo et al., 2023, 2021, 2025b), and cues (Bordalo et al., 2025a, 2026, 2025b; Conlon and Kwon, 2025; Gennaioli and Shleifer, 2010). We differ from this literature in our focus on how selective memory and cues affect the *supply* of information. Our findings are consistent with work in psychology demonstrating that, in fluent processing modes, reliability is rarely scrutinized, with “epistemic vigilance” being triggered only by specific cues such as inconsistency or explicit uncertainty (Kahneman, 2011; Sperber et al., 2010). They also connect to Stanovich (2018) and others’ framework of “miserly” processing, according to which people substitute easy answers for hard ones to save cognitive effort. Overcoming these cognitive shortcuts requires two things: recognizing when the easy answer is insufficient and the problem demands more careful thought, and having the mental tools to produce a more sophisticated answer in that case. In our case, the bottleneck appears to lie in the former step: reliability information is lost not because transmitters lack the capacity to incorporate it, but because it fails to come to mind unless explicitly cued.

Our paper builds on a large literature on social learning (Banerjee, 1992; Bikhchandani et al., 1992; Bursztyn et al., 2014; Conlon et al., 2025; Golub and Sadler, 2016; Golub and Jackson, 2010; Mobius and Rosenblat, 2014; Weizsäcker, 2010), information diffusion (Akçay and Hirshleifer, forthcoming; Banerjee et al., 2013, 2019; Han et al., forthcoming), face-to-face interactions (Atkin et al., 2022; Battiston et al., 2021), and verbalization (Batista et al., 2024). We differ from this literature in our focus on the transmission of qualitative information in the form of spoken narratives and the investigation of cognitive mechanisms that shape transmission of different kinds of information.²

2 Baseline Experimental Design

Our baseline design comprises two experiments. In the transmitter experiment, respondents listen to a recording and are incentivized to pass on the information contained in the recording. In the listener experiment, different respondents listen to either the original recordings or transmitted versions before forming their beliefs.³

Our baseline study design is guided by the following objectives: (i) an experimental setting in which we can quantify the transmission rates of different kinds of information in natural-language spoken messages, (ii) well-defined incentives for transmission, (iii) systematic variation in different types of information in the original recordings, and (iv) an incentive-compatible belief elicitation in the listener experiment to quantify information loss due to transmission.

²Transmission experiments are also studied outside economics (Carlson, 2018, 2019; Mesoudi and Whiten, 2008), reaching at least back to Bartlett’s 1932 studies on *serial reproduction* of stories from memory (Bartlett, 1995).

³The full set of experimental instructions for all experiments can be found at the following link: https://raw.githubusercontent.com/cproth/papers/master/LiT_instructions.pdf.

2.1 Transmitter Experiment

Structure of the experiment. In the transmitter experiment, respondents listen to one recording containing two separate opinions about two economic variables, in a random order: home price growth in an anonymous U.S. city and revenue growth of an anonymous U.S. retail company. The city and retailer are New York City and Walmart, respectively, which is not revealed to participants so that they lack strong priors and cannot search for additional information. This ensures that belief formation is, as much as possible, based only on the information we provide in the original recordings. The opinions are written and recorded by us; respondents are informed that these opinions are based on real media commentary on these topics, and are told at the end of the survey that other participants heard recordings arguing for the opposite conclusions. The recording containing both opinions lasts for 2-3 minutes, with each opinion lasting 1-1.5 minutes.⁴ Respondents are then asked to separately record their own verbalizations of the two opinions they listened to, and finally answer several belief questions about each topic. Appendix Figure A1 shows the structure of the transmitter survey.

Speech recordings. We collect audio recordings, which have several advantages over written text for our purposes. First, oral information transmission is natural: it is the dominant form of communication in daily life, and an important source of information through conversations as well as consumption of television, radio, or podcasts. Second, unlike written communication, the spontaneity of oral communication provides a testing ground for analyzing how cognitive constraints affect information transmission and social learning. A vast literature has examined differences between written and spoken text production (e.g., Akinnaso, 1982; Berger and Iyengar, 2013; Chafe and Tannen, 1987). Written text tends to be more formal, structured, pre-meditated, and involves higher cognitive effort (e.g., Bourdin and Fayol, 2002). Third, speech data allow us to capture critical features of natural language that are mostly absent from written texts and may be essential to the communication of reliability, including tone, emphasis, and disfluencies such as pauses, repetitions, revisions, hesitations, or filler words.

Transmitter incentives. The design of our baseline transmitter incentives directly follows our conceptualization of a message’s information content as *the average belief movement induced by that message*. For each topic, transmitters are tasked with recording a message that induces belief changes that are as close as possible to the average belief changes induced by the original message they listened to. Specifically, one in ten transmitters is selected to be eligible for a \$20 bonus payment. Their probability of receiving the payment (conditional on eligibility) is a quadratic function of the distance between the average belief change induced by their message and the average belief change induced by the original message, among two sets of listeners who will hear either their message or the original message. We explain to respondents that in order

⁴We provide people with the two forecasts consecutively in the same recording, rather than separately playing each forecast before the respondent records their verbalization of it, because this mimics an aspect of transmission in the real world: people are, over time, exposed to multiple pieces of information on various topics, before eventually relaying some information to others.

to maximize their chances of receiving the bonus, they should pass on anything from the original message that they think would be relevant for how people change their beliefs.

This incentive scheme is motivated by our conceptualization of information content and is thus the natural starting point for our experiments. However, there are many alternative possible schemes, some of which may seem less complicated and/or more explicit. Three remarks are in order. First, the optimal strategy under this incentive scheme is to perfectly replicate the original message. Insofar as cognitive constraints prevent this, the optimal strategy is to equate the marginal returns to transmitting level and reliability information by balancing the relative marginal costs of transmitting each type against the relative marginal impact of each type on listeners' belief updates. We study the actual impacts of each type of information on belief updates in Section 3 and transmitters' beliefs about the costs and benefits of transmitting each type in Section 4. Transmitters' beliefs may be biased, which would be a source of transmission distortions that we would want to capture.

Second, incentives based on listeners' *belief changes* (rather than *posteriors*) incentivize transmission of all relevant pieces of information in the original message. If transmitters were incentivized by the accuracy of listeners' posteriors, the optimal strategy might be to "do the updating for the listener:" form a Bayesian posterior after listening to the original recording and simply report this quantitative posterior in the transmitted message. Because transmitters do not know listeners' priors or how their beliefs might react to different pieces of information, incentives based on belief changes encourage them to pass on all information in the original message.⁵ We consider this a naturalistic feature of our scheme: in practice, people most often transmit information without knowing which aspects of the original information the audience wants to learn about and what their priors are, motivating transmission of the substantive information content.

Third, although the quantitative formula underlying the incentive scheme is complicated, we explain the scheme in intuitive terms ("you should pass on all information you think is relevant to how people change their beliefs"). To ensure high levels of understanding, only participants who pass a comprehension question on transmitter incentives are allowed to take part in our study. In Section 4, we explore alternative transmission incentive schemes.

Structure of original recordings. The original recordings have the following general structure. First, they introduce the variable of interest, i.e., home price growth or revenue growth of a retailer. They then put forward some arguments justifying why the variable of interest will increase or decrease. For example, the speaker mentions that as consumers' disposable incomes decrease due to inflation, they often switch towards lower-price retailers, such as the U.S. retailer in question; or that issuance of new residential construction permits in the U.S. city being discussed has slowed down recently, meaning housing supply will increasingly fall behind growing demand. Towards the end of the message, the speaker states explicitly whether they believe the variable will increase or decrease over the coming year. Throughout the recording, the reliability

⁵Even under our incentive scheme, rational transmitters might, instead of passing on the original information, communicate the degree of belief movement they think should occur given their assumed distribution of prior beliefs, updating rules, etc. In practice, this behavior appears negligible: we observe no transmitter recordings suggesting attempts to convey predicted belief movements.

of the prediction is explicitly or implicitly communicated using techniques we discuss below. Full transcripts of the messages and the audio recordings are available in Appendix F.

The design of these messages is motivated by the nature of real-world commentary on economic topics such as house price or company revenue growth. Such commentary usually justifies predictions with substantive arguments about the variables of interest, e.g., relating to market conditions or broader trends in the economy. The arguments in our messages are drawn from real media reporting on these topics. Moreover, such messages communicate reliability with both explicit and implicit markers.

Experimental variation: original recording contents. The design of our original recordings is guided by our distinction between the *level* and *reliability* of a prediction about a variable. We make the following observations about this distinction. First, this distinction is parsimonious, theoretically appealing, and general. To perform a belief update from any piece of information, a Bayesian agent always requires both a signal value and a signal precision. Moreover, level and reliability are always—implicitly or explicitly—conveyed by any forecast. For example, even the absence of explicit confidence or reliability statements could itself be an indicator of the forecast’s reliability. Second, our distinction connects with previous belief formation research: for example, some research suggests that people pay insufficient attention to the *weight* or precision of evidence when forming their beliefs in abstract and quantitative updating tasks (Griffin and Tversky, 1992; Massey and Wu, 2005). Third, note that our taxonomy is different from the distinction between the first and second moment of the forecast state. Specifically, reliability is an attribute of a signal structure rather than a property of the distribution of the forecast state.

To leverage the level-reliability distinction in our experiments, we randomize these two features of the original message recordings: whether the message argues for an increase or a decrease in the level of the variable and whether the message is reliable or unreliable.

We randomly assign respondents to two kinds of reliability manipulations. Respondents in the *naturalistic condition* hear recordings that vary reliability using a combination of explicit statements about confidence, evidence quality, and speaker competence, as well as implicit markers of reliability such as verbal fluency and vocabulary. For example, a high-reliability message sounds highly fluent with a sophisticated vocabulary, cites respectable sources of evidence, and mentions relevant credentials. A low-reliability message is full of disfluencies, expresses low confidence, cites obviously unreliable sources, and admits a lack of relevant credentials.

Meanwhile, respondents in the *modular condition* receive recordings that are identical except for a set of explicit markers indicating either high or low reliability (e.g., “definitely” vs. “possibly”, “will” vs. “might”, etc.) and explicit confidence statements (“I am highly confident” vs. “I am not at all confident”). Respondents in the *modular condition* are assigned to one of the following three conditions: (i) Strong reliability; (ii) Weak reliability; and (iii) Neutral reliability (where the markers and confidence statements are simply omitted).⁶

These two types of manipulations serve different purposes: the *naturalistic condition* em-

⁶As pre-specified, our main analysis focuses on comparisons between weak and strong reliability for simplicity. Appendix Figure A4 shows belief updates including the neutral-reliability condition.

braces the full range of linguistic tools through which reliability of a statement may be expressed in practice, at the cost of a loss of control about which precise component drives perceptions of reliability. The *modular condition*, by contrast, provides this control by allowing us to trace the loss of specific reliability words or phrases, at the cost of focusing attention on just these modular elements. While interpretation of verbal uncertainty prefixes can vary from person to person (Vogel et al., 2022), this variability is constant between listeners to original and transmitted messages, meaning tracking the loss of these words should suffice to decompose information loss. Because both manipulations end up producing very similar results, we report our main results pooling both conditions, and show disaggregated results in Appendix Figure A7.

Our reliability manipulations most closely approximate real-world situations where a person is learning from a stranger, about whose reliability they have no strong prior. In these cases, people infer a speaker’s reliability from the way the speaker talks, the claims the speaker makes, and what the speaker says about their background. All of the participants in our experiment are strangers to each other and must infer reliability only from the contents of voice recordings. Situations like this abound in everyday life, in contexts such as social media, television, conferences, public venues, social gatherings or professional settings.

Finally, we randomize whether the recording is made by a male or female actor. This is not a focus of analysis and we randomize simply for symmetry, and so that each topic a transmitter listens to is discussed by a different voice. We find no evidence that the effects of any of our manipulations, or the effects of transmission, vary with the original voice’s gender.

The different margins of randomization are stratified: each transmitter hears two recordings, one with an “increase” and one with a “decrease,” one with “strong reliability” and one with “weak reliability,” and one with a male voice and one with a female voice.⁷

Beliefs. After recording themselves, transmitters answer the same beliefs questions that listeners do, so we defer discussion of those questions to the following subsection.

2.2 Listener Experiment

Structure and treatments. This experiment draws on the speech recordings collected in the transmitter experiment. It lets us quantify transmission-induced information distortions by measuring and comparing the information content of the original messages and transmitted versions of those messages.

Recall that our experiments involve forecasts about two topics: (i) the change in home price growth in a U.S. city and (ii) the change in revenue growth of a U.S. retailer, both for the upcoming year. For each of the two topics, participants in the listener survey first state their prior belief about the outcome variable of interest and then listen to a recording about the variable before answering a set of beliefs questions. The order of the topics is randomized. For each topic, respondents are randomly matched to a transmitter and listen either to the same original

⁷Then, if exactly one of the two topics is in the modular condition, that topic has a 33% chance of getting switched to “neutral reliability”. If both topics are in the modular condition, there is a 66% chance that one of the two topics is randomly switched to “neutral reliability.”

recording as the transmitter heard, or that transmitter’s message. We implement a 30% chance of hearing the original and a 70% chance of hearing a transmitted recording. We oversample transmitted recordings as they are by construction more heterogeneous compared to original recordings. Appendix Figure A2 shows the survey structure.

Listeners are told whether they are listening to the original message or another participant’s transmitted version. They could take this information into account when updating their beliefs about the message content, e.g., by discounting the reliability of *any* transmitted message relative to a corresponding original message. However, as discussed in Section 3, we find no evidence that transmission on average affects the perceived level or reliability of the original messages.

Beliefs. After listening to a recording, respondents are incentivized to guess the realization of the target variable—change in house price growth or change in revenue growth over the next 12 months—as well as the level and reliability of the prediction in the original message.

We separately elicit beliefs about the state of the variable under discussion, referred to as *state beliefs* henceforth, as well as beliefs about the original message’s contents, called *message beliefs*, for two reasons. A listener’s state beliefs are the most economically relevant objects. However, belief movements about the state are also affected by respondents’ priors and prior confidence, making it difficult to back out respondents’ perceptions of the level and reliability of the original prediction. Directly eliciting beliefs about the message’s level and reliability circumvents this issue and brings us closer to the objects of interest in our guiding distinction and our treatment manipulations. Moreover, belief updates about the state are *simultaneously determined* by a message’s level and reliability. This means that loss of level information affects respondents’ sensitivity to reliability information and vice versa, preventing us from cleanly distinguishing level and reliability information loss based solely on state belief updates. The same is not true for message beliefs, which separate out the original message’s level and reliability.

For each topic, we hence elicit three outcome variables: state belief movements (posteriors about the economic variable minus priors); and two message beliefs: the respondent’s belief about the level of the original message’s prediction and the respondent’s belief about its reliability. We elicit respondents’ priors about the state, and use belief movements as our outcome rather than posteriors. This reduces noise resulting from idiosyncratic respondent heterogeneity in usage of the belief scales. Because respondents do not know anything about the context the message concerns, their priors simply capture tendencies in their response behavior.

To measure respondents’ state beliefs we ask them about the change of the variables of interest in the next 12 months. For home price growth, this question reads as:

How will house price growth in this city change over the next 12 months?

Our two unknown states are *changes in growth rates* because this permits a natural prior of zero and reasonably symmetric possibilities around that prior. This lets us shift beliefs symmetrically up or down with our high- or low-level messages, creating clean variation in the information content of the recordings. To elicit respondents’ corresponding message beliefs about the level of the prediction, we ask the following question:

How do you think the person [whose opinion you just heard/whose opinion was summarized in the recording] predicts house price growth in this city will change over the next 12 months?

To measure respondents' message beliefs about the reliability of the prediction, we ask:

How reliable do you think the prediction given by the person [whose opinion you just heard/whose opinion was summarized in the recording] is? Specifically, what do you think is the probability that this person's forecasts about changes in house price growth in this city are roughly correct? Concretely, assuming that the true change in house price growth is a number called X , what do you think is the likelihood that this person's prediction will fall within 1% of X , i.e., between $X-1\%$ and $X+1\%$?

Incentives for accuracy. Respondents are told that one in ten respondents will be randomly chosen to be eligible for a \$20 bonus payment, which will be based on one of the incentivized items in the survey. State beliefs are always directly incentivized based on the true development of the variable over the next year.⁸ Message beliefs are unincentivized for a randomly selected 50% of respondents. For the other half of respondents, the question is phrased as a second-order question ("your job is to predict what people who heard the same recording as you would on average respond to the direct question") and responses are incentivized based on the accuracy of their guess about other participants' average guess.⁹ Results based on incentivized versus unincentivized message beliefs are virtually identical, as shown in Appendix Figure A8.

2.3 Sample and Procedures

We conducted our experiments on Prolific, a widely used online platform to conduct social science experiments (Eyal et al., 2021). The transmitter experiment and listener experiment were run with 540 and 1,510 U.S. respondents, respectively, in November 2023. Table A3 records summary statistics for all our experimental samples. All of the data collections were preregistered on the AEA RCT registry (#12119, #17479, #17533, #17540). As preregistered, we drop recordings below the 5th percentile of recording length or transcript word length (as a proxy for empty or contentless recordings). Following this restriction, our baseline transmitter experiment yields a total of 1,010 valid speech recordings.

3 What Is Lost in Transmission?

Our main finding in this paper is *differential information loss*: information about the reliability of a forecast is lost in transmission much more strongly than information about its level. In this

⁸State beliefs are incentivized with the following formula: Probability of winning \$20 [in %] = $100 - 10(\text{Estimate [in \%]} - \text{True state of the world in 12 months [in \%]})^2$.

⁹Responses are incentivized with the following formula for beliefs about the originator's prediction and reliability, respectively: Probability of winning \$20 [in %] = $100 - \alpha(\text{Response [in \%]} - \text{Average response to direct question [in \%]})^2$, where $\alpha = 10$ for level and $\alpha = 2$ for reliability. This approach allows us to incentivize these beliefs in the absence of a "true state", since the original recordings were provided by us and there is no corresponding originator belief.

section, we demonstrate differential information loss in three distinct and complementary ways: first, by examining the transcripts of transmitted messages for mentions of level and reliability; second, by analyzing listeners’ *message beliefs* about the level and reliability of the original forecast, separately for listeners who hear original messages versus listeners who hear transmitted versions; and third, by analyzing listeners’ *state beliefs* about the economic variables discussed in the original forecasts. Each set of analyses has its own advantages and drawbacks.

3.1 Transcript Analysis

We begin by examining the transcripts of transmitted messages to see whether they contain statements about the level or reliability of the original forecasts.

Panel (a) of Figure 1 displays the share of transmitter transcripts classified as containing statements about the level of the original prediction or about its reliability. For reliability, we adopt a maximally broad notion of what counts as communicating reliability, incorporating all of the components we use to vary reliability in the original recordings. This includes explicit statements about reliability or confidence as well as the use of certainty or uncertainty markers like “might” or “definitely.” We separately show results of human coding and of automated coding using the large language model GPT-4 (Asirvatham et al., 2026). The figure illustrates that the different coders and the large language model come to similar conclusions.¹⁰ See Appendix Table A2 for some examples of transmitted messages and their hand-codings.

The key finding of Panel (a) is that while most transmitted scripts contain statements about the level (between 87 and 95 percent), a far smaller fraction of transmitted scripts contain statements indicating the reliability of the original message (between 30 and 45 percent). Panel (b) shows that this is true independent of the length of the transmitted message: even among transmitted messages that are 200-300 words long (longer than the original messages), only 20% are unanimously agreed upon by our coders to contain statements about reliability. Longer messages tend to differ from shorter ones primarily in providing a much higher level of detail about the original message’s arguments for its level prediction. Appendix Figure A10 also shows that the fraction of scripts containing statements about level or reliability is fairly stable across our four level $\times$ reliability conditions (high versus low level, and weak versus strong reliability).

Transmitted recordings also include many disfluencies—hesitations, “um” statements, self-corrections, and so on—which could influence how listeners perceive the reliability of the original forecast. However, Appendix Figure A11 shows that the average number of disfluencies in transmitted scripts does not vary with whether the original forecast was high- or low-reliability, and the presence of disfluencies in transmitted scripts does not affect listener beliefs about the reliability of the original forecast (analyzed in the next section). The reliability of the original forecasts does not appear to be effectively communicated through disfluencies.

¹⁰For level, if one human coder identifies level as being passed on, the other does with 91% probability, and GPT does with 98% probability. For reliability, the corresponding numbers are 60% and 75%. In our analysis of beliefs data where we split according to handcoded classifications, we restrict to transcripts where our coders agree unanimously.

Result 1. *Transmitted messages are about three times more likely to mention the level of the original prediction than to mention its reliability.*

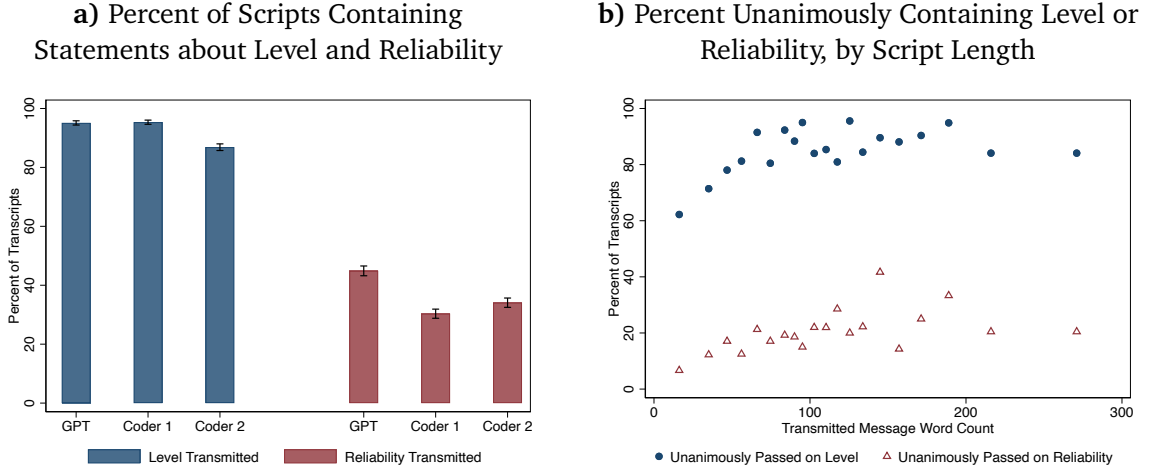


Figure 1: This figure presents data from our baseline experiment (Belief Movement Incentives). Panel (a) shows the percent of transcripts that are coded as conveying any information about the level and reliability of the original forecast, separately by two human coders and GPT-4, with standard error bars. Panel (b) shows binned scatterplots of the percentage of scripts unanimously classified by our coders and GPT as containing statements about level or reliability, respectively, against the word count of the script. $N = 540$ transmitters, 1,010 transcripts.

These results, while transparent and strongly suggestive, do not conclusively establish that reliability information is lost in transmission more than level information. For example, perhaps listeners only need to hear 10% of the reliability cues in the original messages to grasp their true reliability, so that 90% of the original cues can be dropped without loss; or perhaps reliability is passed on in a way not picked up by our codings. To address this possibility, we turn to an analysis of listeners’ *message beliefs*, their beliefs about the level and reliability of the original messages. We will confirm our finding of differential reliability loss, but also find that our binary hand-codings from this section do not tell the full story: there is substantial loss of reliability information even among messages we coded in this section as passing on at least some reliability.

3.2 Message Beliefs

To provide independent measures of level and reliability information, we separately elicit listeners’ message beliefs about the level and reliability of the original prediction, using questions described in Section 2.2. Figure 2 presents results on message beliefs. We z-score message beliefs within each topic $\times$ reliability manipulation quadrant to make the aggregation across experimental conditions more comparable; results with raw beliefs are available in Appendix Figure A3.

Panel (a) examines message beliefs about the level of the original prediction. The blue dots show the average beliefs of listeners who directly hear original recordings. Listeners who hear a *low*-level original recording believe the level of the prediction is 1.37 SDs lower on average than listeners who hear a *high*-level original recording. Meanwhile, the orange dots show the beliefs of listeners who hear transmitted versions of the original recordings. Here, the difference

between the beliefs of listeners who hear transmitted versions of *low*-level recordings and those who hear transmitted versions of *high*-level recordings is only 0.88 SDs, indicating $100 \times [(1.37 - 0.88)/1.37] \approx 35\%$ loss of sensitivity to level information. In other words, listeners who hear transmitted recordings are 35% less sensitive to variations in the level of the original predictions, compared to listeners who directly hear the original predictions. Formally, the *change in slope* statistic printed in the plot is calculated from a regression of the form

$$\text{LevelBelief}_i = \beta_0 + \beta_1 \text{HighLevel}_i + \beta_2 \text{Transmitted}_i + \beta_3 (\text{HighLevel}_i \times \text{Transmitted}_i) + \varepsilon_i, \quad (1)$$

where LevelBelief_i is the listener’s belief about the level of the original prediction (z-scored at the topic by reliability manipulation type level); HighLevel_i is a dummy for the original forecast having a high level; and Transmitted_i is a dummy for the participant listening to a transmitted version of the original forecast. Standard errors are two-way clustered at the voice recording and listener level.¹¹ The change in slope statistic is simply $-100 \times (\beta_3/\beta_1)$.

Panel (b) examines listeners’ message beliefs about the reliability of the original predictions. Here, the sensitivity loss is nearly three times as strong. Listeners hearing the original messages believe the strong-reliability messages are 1.18 SDs more reliable than the weak-reliability messages on average. Among listeners hearing transmitted versions of the original messages, this difference is only 0.12 SDs, indicating roughly 90% loss of sensitivity. A formal test of equality of the two information loss statistics rejects the null at $p < 0.001$, $\chi^2 = 74.5$.

Figure 2 further illustrates that, in both cases above, transmission weakens the distinction between high- and low-level messages (or weak- and strong-reliability messages) by *symmetrically compressing listeners’ beliefs towards an intermediate value*. This is compatible with the following dynamic: listeners hold an average prior about level or reliability that is located halfway between our two manipulations; they update away from this prior when hearing a message; and the strength of this update is weakened by noise introduced during transmission. If more noise is introduced for the reliability than the level, this compression will be stronger for reliability.

The finding of nearly symmetrical compression also shows that, contrary to an intuitive hypothesis, the fact that a message is transmitted does *not* reduce its perceived reliability on average: instead, transmission causes strong-reliability messages to be perceived as less reliable, and weak-reliability messages to be perceived as more reliable.

Interpretation. The finding that transmission symmetrically compresses level and reliability beliefs admits a cognitive-imprecision interpretation according to which transmission adds symmetric, zero-mean noise to the level and reliability of the original message, attenuating listeners’ perceptions of those parameters towards a default belief as the true parameters become difficult to discern. This interpretation connects our findings to noisy processing models popular in the recent literature (e.g., Ba et al., 2024; Enke and Graeber, 2023). Alternatively, transmission-induced compression towards an intermediate belief could reflect a form of ignorance or feeling of “I don’t know” (Fischhoff and Bruine De Bruin, 1999), or a process of anchoring-and-adjustment (Tversky and Kahneman, 1974) caused by listening to a transmitted message. This

¹¹Standard errors are virtually identical for different ways of clustering.

could be true if listeners find it difficult to decode or interpret the contents of transmitted messages, give up, and retreat to a default belief. Under this interpretation, the strength of attenuation reflects the difficulty of decoding each type of information. We remain agnostic about which exact interpretation is the most accurate.

Result 2. *Verbal transmission induces substantial information loss. This information loss differs for different types of information: Loss of reliability information is about three times as large as loss of level information.*

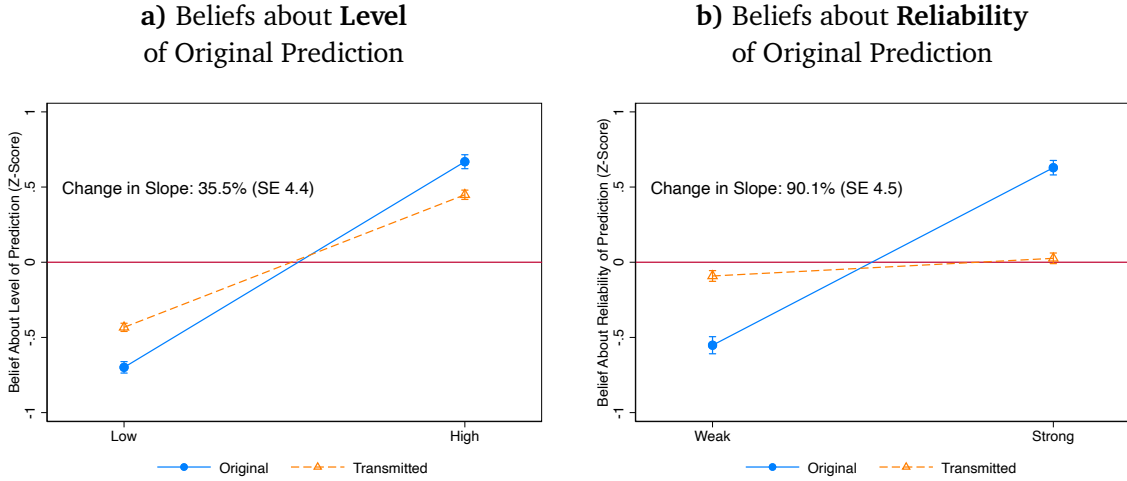


Figure 2: Slopes in the two panels differ at $p < 0.001$, $\chi^2 = 74.5$. This figure presents data from our baseline experiment (Belief Movement Incentives). It shows listeners’ beliefs about the level and reliability of the prediction in the original message, separately by whether the original message is low- vs high-level or weak- vs strong-reliability, and separately by whether the listener hears the original message or a transmitted version of it. Dots are mean beliefs (z-scored at the topic by reliability manipulation level) and bars are standard error bars (1 SE each direction). See Appendix Figure A3 for results with raw (non-z-scored) beliefs and Tables A5 and A6 for regression tables. $N = 1,510$ listeners and 540 transmitters.

Relation to transcript analysis. How much of the differential information loss in Figure 2 simply reflects the complete omission of reliability indicators from two-thirds of transcripts documented in the previous section? Appendix Section D decomposes differential information loss into a component driven by the absence of reliability indicators (extensive margin) and a component that remains even in messages with at least one reliability indicator (intensive margin). We find that only 30-50% of the differential loss in Figure 2 is driven by the extensive margin; as we show, even messages that contain at least one reliability cue drop many of the cues seeded in the original messages, and this matters for listeners’ beliefs.

3.3 State Belief Updates

Message beliefs, analyzed above, enable us to separately track level and reliability information through the transmission process. However, they are artificial objects. The objects of immediate

economic relevance are listeners' *state belief updates*, their belief updates about the economic variables discussed in the recordings (home price growth and revenue growth).

Differential information loss in state belief updates. We can adapt the specification in Equation 1 to measure level and reliability information loss using state belief updates instead of message beliefs. The results of this analysis are printed in Panel (c) of Figure 3. We calculate that transmission reduces the sensitivity of listeners' belief updates to our level manipulations by 30%, and reduces the sensitivity to reliability manipulations by 90%, strikingly similar to the numbers calculated using message beliefs.

Informally, Panel (c) of Figure 3 shows that listeners of the original messages update, on average, twice as much from strong-reliability messages compared to weak-reliability messages. Listeners to transmitted versions, meanwhile, update almost the same amount from weak- and strong-reliability messages. This is what underlies our finding of 90% reliability information loss.

Interpreting the effects of transmission loss on state belief updates. In addition to showing strong differential loss of reliability information, Figure 3 displays a rich set of patterns that we now discuss. Panel (a) displays the average state belief updates of listeners who directly hear original recordings, across the four categories of our level/reliability cross-randomization. We pool data from both topics, home price and revenue growth, and separately z-score belief movements for comparability. The panel shows that state belief updates are sensitive to both our level and reliability manipulations. In particular, listeners adjust their beliefs in a qualitatively Bayesian manner: they move in the direction of the forecast they receive, with the strength of the update moderated by the reliability of the forecast.

Panel (b) of Figure 3 shows predictions for the effects of transmission loss on state belief updates. Panel (c) shows the actual effects of transmission, which match the predicted effects.

To understand the predictions and results, observe that the loss of *level* information should uniformly shrink listeners' belief updates towards zero (the mean belief update, given z-scoring). This is because, as Panel (a) of Figure 2 shows, transmission symmetrically compresses beliefs about the level of the original prediction towards the mean value. This should in turn compress belief updates towards the mean belief update, given that average priors are the same across experimental conditions. Hence, across all four conditions, we predict that level information loss should attenuate belief updates towards zero (the green arrows in Panel (b) of Figure 3).

Meanwhile, the loss of *reliability* information should have different effects in the strong versus weak reliability conditions. Loss of reliability information symmetrically compresses listeners' beliefs about the reliability of the original messages towards the mean (Panel (b) of Figure 2). This means that transmission causes *strong-reliability* messages to be perceived as *less reliable*. This, in turn, should shrink belief updates from strong-reliability messages, since the size of a listener's belief update should be smaller the lower the perceived reliability of the signal. Hence we predict that in the strong-reliability conditions, reliability information loss should attenuate belief updates towards zero (the purple arrows in the leftmost and rightmost conditions in Panel (b) of Figure 3). Conversely, transmission causes *weak-reliability* messages to be perceived

as *more reliable*. This means reliability information loss should strengthen belief updates *away* from zero in the weak-reliability conditions (the purple arrows in the two middle conditions).

Overall, we obtain an unambiguous prediction that in the strong-reliability conditions—where both level and reliability information loss push in the same direction—transmission should cause belief updates to shrink strongly towards zero. Meanwhile, in the weak-reliability conditions, level information loss pushes towards zero and reliability loss pushes away from zero; without knowing which effect dominates, we have an ambiguous prediction for the effect of transmission belief updates in these conditions.

Panel (c) of Figure 3 shows empirical results that exactly bear out these predictions. In the strong-reliability conditions, transmission causes average belief updates to shrink in size by about 50%. Meanwhile, in the weak-reliability conditions, the opposing effects of level and reliability information loss seem to roughly cancel out, and average belief updates barely change.

Appendix Figure A5 validates our comparative-static explanation of the empirical results by splitting transmitters according to whether they are coded as passing on reliability (see Section 3.1). Consistent with our story, transmitters who fail to pass on reliability information induce overreactions among listeners in the weak-reliability buckets and more severe underreactions among listeners in the strong-reliability buckets.

Implications of transmission loss. Summing up, Figure 3 shows that transmission-induced information loss has two impacts on downstream belief updates. First, listeners to original messages update about twice as much from strong-reliability messages as from weak-reliability messages; by contrast, listeners to transmitted versions update to the same degree from weak- and strong-reliability messages. That is, transmission *reduces the relative influence of strong-reliability beliefs*. This happens mainly through a reduction in the absolute influence of strong-reliability messages; the absolute influence of low-reliability messages does not change much on average because reliability loss is offset by level information loss pushing in the opposite direction. This suggests that the main impact of transmission loss is to reduce the impact of information from trustworthy sources. Second, averaging across all four conditions, listeners’ absolute belief updates are 30% smaller when listening to transmitted messages, an effect driven by the strong-reliability conditions.¹² This means that transmission *reduces the average impact of new information on beliefs*. Whether transmission-induced information loss leads to more convergence or more polarization of beliefs in response to new information is more subtle; it depends on the cueing process underlying information loss. We demonstrate and discuss this in Appendix C.

Result 3. *Verbal transmission weakens the average effect of new information on beliefs. It also reduces the relative influence of strong-reliability, compared to weak-reliability, information.*

¹²Technically, the figure shows that *z-scored* belief updates are smaller, but this is also true for mean raw belief updates; the mean raw belief update is ≈ 0 .

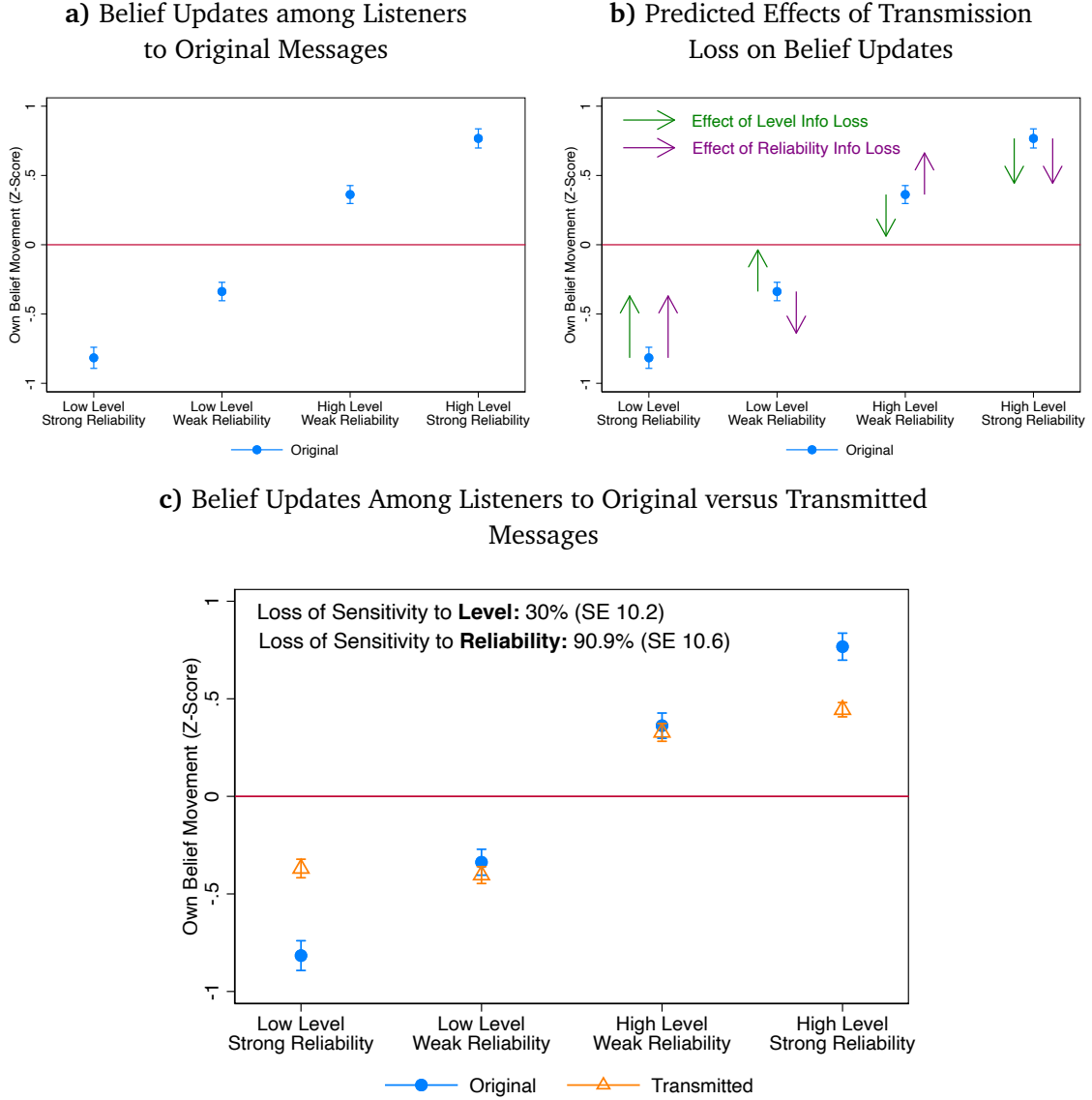


Figure 3: This figure shows average belief movements (posterior minus prior) about the economic variable from our baseline experiment (Belief Movement Incentives). Panel (a) shows average belief movements about the economic variable across the four different level-reliability conditions, only for listeners who directly hear the original messages. Dots are mean beliefs and bars are standard error bars (1 SE each direction). Panel (b) adds illustrative arrows. Panel (c) adds the corresponding beliefs of listeners hearing transmitted versions of the messages. $N = 1,510$ listeners and 540 transmitters. The loss of sensitivity to level information is calculated from a regression of the form: $\text{BeliefUpdate}_i = \alpha_0 + \alpha_1 \text{HighLevel}_i + \alpha_2 \text{StrongReliability}_i + \alpha_3 \text{Transmitted}_i + \alpha_4 (\text{HighLevel}_i \times \text{Transmitted}_i) + \xi_i$. The loss of sensitivity to reliability information is calculated from a regression of the form $\text{BeliefUpdate}_i \times (2 \times \text{HighLevel}_i - 1) = \gamma_0 + \gamma_1 \text{HighLevel}_i + \gamma_2 \text{StrongReliability}_i + \gamma_3 \text{Transmitted}_i + \gamma_4 (\text{HighLevel}_i \times \text{Transmitted}_i) + \zeta_i$, where we flip the sign of low-level belief updates to make the effects of StrongReliability comparable across low- and high-level messages. Appendix Table A4 gives regression versions of these results. Figure A4 shows these results restricting to the Modular manipulation and including the neutral-reliability condition. Appendix Figure A5 shows empirical results validating the arrows in Panel (b). Appendix Figure A3 shows raw (non-z-scored) beliefs. Appendix Figure A6 shows Wasserstein distances between the belief update distributions.

4 Mechanisms Underlying Differential Loss

What drives the differential loss of reliability and level information? In this section, we test different potential mechanisms. To structure this analysis, we distinguish between mechanisms that involve a deliberate decision by the transmitters to prioritize passing on level information, and mechanisms that involve transmitters subconsciously or non-deliberately failing to pass on reliability information. If differential loss results from transmitters' deliberate decisions, it arises either because (i) the *perceived benefits* of transmitting reliability information are lower or (ii) the *perceived costs* of transmitting reliability information are higher. Alternatively, differential loss may not result from a deliberate cost-benefit tradeoff; for example, (iii) reliability information may simply *fail to come to mind* at the moment of recording the voice message, e.g., due to some kind of attention or memory constraint. We examine each of these three possibilities in turn.

4.1 Perceived Benefits of Transmitting Level and Reliability

We first consider the perceived benefits of, or incentives for, communicating level versus reliability information. Perceived incentives are a natural starting point: in practice, people pass on information in a variety of different circumstances, and the objective of such information transmission can vary widely, from informing to persuading to entertaining the recipient. It is likely that people (at least partly) tailor the contents they transmit to the specific requirements of the situation. The differential loss observed in our data might be an artifact of our setup that induces specific (perceived) transmission incentives, or it may be a more fundamental property of transmission that is likely to occur robustly across different transmission settings.

4.1.1 Evidence from Baseline Experiment

We begin by examining three additional pieces of evidence from our baseline experiment. After recording their messages, participants in our transmitter survey are randomized between three sets of supplementary questions. First, we elicit beliefs about the importance of passing on reliability versus level information, to test whether participants (mistakenly) believe that passing on reliability information would not affect listeners' belief updates and hence their probability of receiving the bonus payment. To the contrary, we find that respondents believe that passing on reliability information increases their probability of receiving the bonus payment by about the same amount as passing on level information: the average response is 71% for level and 68% for reliability. This is true even among respondents whom we classify as *not* passing on reliability information in their recordings (averages of 73% versus 66%).

Second, to test whether respondents are aware that they are omitting specific information, we ask another one-third of respondents whether they included level information and reliability information in their recordings. In line with our analyses of transcripts, we find that 64% of respondents admit to not passing on reliability information (compared to 31% for level).

Third, to examine whether people forget or do not pay attention to the incentive scheme, we examine whether, at the end of the survey, the final one-third of respondents still pass the initial

comprehension checks about their incentives. We find that 90% of respondents correctly answer both questions about the incentives.

Taken together, these results suggest that — when asked — people remember the incentive scheme and believe it incentivizes them to pass on reliability, yet they admit to not passing on reliability in their actual recordings. This provides a first sign that the loss of reliability information is due to the failure of reliability to come to mind during the transmission process.

4.1.2 Additional Evidence: Incentives for Content Transmission

In the baseline experiment, transmitter bonuses were based on the induced belief movements of listeners, leaving transmitters free to pick and choose which dimensions of the original content they believe will be relevant for listeners’ belief updates. In a supplementary experiment, we instead explicitly fix the relative incentives to transmit level and reliability information, making beliefs about the relative importance of level and reliability information for belief updates irrelevant for transmission decisions.

This experiment and its results are described in detail in Appendix Section A.1.3. Half of the participants are told that their message will be scored by a group of “evaluators” on a scale of 0 to 10 according to the fidelity with which it preserves the “content and meaning” of the original message. Compared to our baseline incentives, this scheme should incentivize transmitters to pass on reliability information regardless of their beliefs about its importance for listeners’ belief updates, because the instructions encompass *all* the contents of the original message. The other 50% of transmitters are explicitly told that evaluators will separately score the fidelity of their message’s transmission of the *level* and the *reliability* information in the original message, and that both scores receive equal weight in their probability of receiving the bonus payment. This scheme has two distinctive features. First, it ensures that transmission of reliability information is *explicitly* just as payoff-relevant as transmission of level information. Second, unlike any of our other incentive schemes, it introduces transmitters to the distinction between level and reliability information. It therefore produces a very conservative test of differential information loss.

Results of the experiment are described in Appendix Section A.1.3 and reported in Appendix Figures A15 and A16. We find large differential information loss in both new incentive schemes, though differential loss is slightly smaller than in our baseline experiments. We conclude that differences in perceived importance of transmitting level versus reliability information cannot fully explain our main result.

4.2 Perceived Costs of Transmitting Level and Reliability

Next, we turn our focus to the second possible driver of differential loss and examine the subjectively perceived *costs* or *difficulty* of transmitting level versus reliability information. Here we can distinguish between the (*ex-ante*) *anticipated* and (*ex-post*) *experienced* costs of transmitting each type of information. Transmitters might deliberately omit reliability information because they *expect* it to be more costly or difficult to transmit; alternatively, they might try to transmit reliability information but then *experience* it as being very difficult to properly transmit. Our analysis

in Section 3.1, which found that 60% of transmitted transcripts do not include anything about the reliability of the original message, suggests that transmitters are not even *trying* to transmit reliability, suggesting that anticipated costs are more likely to be relevant than experienced ones.

As a direct test of the initially *anticipated* costs of transmitting level versus reliability information, we conduct an additional transmitter experiment in which we study whether transmitters prefer to be paid for their performance in transmitting information about (i) the level of the original prediction or (ii) the reliability of the original prediction. By “performance,” we mean an external evaluator’s assessment of how well the transmitter’s message passed on the level or reliability information, respectively, following the second incentive scheme from Section 4.1.2. We also elicit transmitters’ expectations about how difficult transmitting level or reliability information will be. To test whether experienced costs deviate from anticipated ones, we study whether transmitters’ beliefs about the difficulty of transmitting level versus reliability information change after experiencing the transmission process.

Panel (a) of Appendix Figure A17 shows that 62 percent of respondents choose to transmit information about the reliability of the prediction, and the average perceived difficulty of transmitting reliability information is slightly lower than for level. Differences in the perceived difficulty of communicating level and reliability information are relatively small, both measured before (Panel (b), $t = 0.64, p = 0.53$) and after the recording (Panel (c), $t = 2.4, p = 0.02$). This suggests that transmitting reliability information is, if anything, *easier*, and makes it hard to see how higher anticipated or experienced costs of transmitting reliability information could play a role in driving our results. Virtually all respondents pass on the characteristic they chose.

There is no heterogeneity in perceived costs that could generate the pattern of differential information loss we observe. For example, suppose that the 60% of people choosing to transmit reliability are capable of transmitting both types of information in the main experiment, but the 40% choosing to transmit level information find transmitting reliability to be prohibitively costly. This could generate differential information loss even if transmitting level is perceived as harder on average. But we find no such heterogeneity: the groups choosing to transmit level versus reliability information give similar average difficulty ratings and have similar average difficulty gaps between the parameter they choose to transmit and the other.

Result 4. *Mechanism experiments suggest that differential transmission loss of reliability information in our experiment is not the result of a deliberate decision: it is not driven by the subjectively perceived benefits or costs of transmission.*

4.3 What Comes to Mind

Having established that differential loss does not appear to be the result of a deliberate prioritization of level information, we examine the possibility that transmitters subconsciously or non-deliberately neglect to include reliability information due to attentional or memory constraints. Our previous results—which show that, when asked, transmitters believe passing on reliability is important, believe it is easier than passing on level, yet mostly do not even try to pass on reliability—hint at this kind of attentional neglect of reliability.

4.3.1 Conceptual Framework

To structure our investigation, we interpret the transmission process through the lens of associative memory models of retrieved context (e.g., Bordalo et al., 2025b; Kahana, 2012). When a transmitter encounters an original message, the message is stored in the transmitter’s memory as a set of traces that include both substantive content (e.g., facts, arguments, and examples contained in the message) and meta-features (the message’s source and credentials). When the transmitter later encounters a cue that triggers transmission—the transmission task, in our experiments, or a question or conversation in real life—the cue activates stored traces with probabilities that increase in the similarity between the cue and the trace and decrease in the interference between the trace and other traces (Bordalo et al., 2024).

In this class of models of episodic memory, the probability of retrieval is affected both by characteristics of the cue (via similarity) and characteristics of the memory database (via interference). This, in turn, suggests two classes of reasons why reliability information might come to mind less than level information in practice; in the rest of this section, we focus on cues.

Cues. The “cue” in our transmission experiments is the generic request to recall and pass on the information from the original message. Whether this cue triggers retrieval of level or reliability information with higher probability depends on the perceived similarity or association between the cue and level or reliability information, respectively.

Anecdotal evidence suggests that real-life requests for transmission typically cue level information but rarely reliability information. People often ask about substantive economic facts, beliefs, or projections, but rarely directly request confidence levels or reliability assessments from their interlocutors. This could create an association between transmission and level information that leads generic transmission contexts (like our experimental context, which carefully avoids explicitly cueing either level or reliability) to naturally cue level information, but not reliability.

Interference. While we will not focus on interference due to the difficulty of measuring it empirically, we note it is plausible that attempts to retrieve reliability face greater interference than attempts to retrieve level information. Predictions about the levels of economic variables are typically supported by, and communicated through, rich contextually-specific claims, arguments, and examples, meaning that different level claims from different contexts are strongly distinct from each other. This leaves little scope for interference between them in the retrieval process. By contrast, reliability is much more likely to be communicated using semantically generic terms such as certainty or uncertainty prefixes, statements about confidence levels, or the name of a source. This creates much more potential for interference between different pieces of reliability information that could block retrieval. Indeed, a longstanding literature on “source monitoring errors” in memory finds that individuals have difficulty accurately attributing particular pieces of memory to a source (Hovland and Weiss, 1951; Johnson, 1997). Several pieces of evidence connect these source monitoring errors to interference between sources in the memory bank (Henkel and Franklin, 1998; Johnson et al., 1993; Lindsay and Johnson, 1989).

4.3.2 Cues and Reliability Transmission

We now provide evidence on the role of cues in reliability transmission. We begin by showing that transmitters remember the reliability of the original message when asked, so reliability information is available for retrieval rather than being irretrievably forgotten. We then provide four pieces of evidence that cues of reliability increase the rate of reliability transmission, ordered from subtle and payoff-irrelevant to salient and payoff-relevant cues. Finally, we show that explicit cues of reliability are rare in practice.

Transmitters remember reliability when cued. Once specifically cued, transmitters in our baseline experiment are able to accurately recall both the level and reliability of the original predictions. Specifically, after they complete their recordings, we present transmitters with the same set of beliefs questions we pose to listeners (see Section 2). Appendix Figure A12 demonstrates that there is virtually no memory loss about the original message’s reliability: several minutes after hearing the original recording and after performing the cognitively demanding task of recording their own voice message in the interim, transmitters are *just as sensitive to variations in reliability* as listeners whose beliefs are elicited immediately after hearing the original recordings. Transmitters hence do not *forget* the reliability information, meaning that, if cued to attend to reliability, they should be able to retrieve and transmit it.

Irrelevant cues in our baseline experiment. An incidental feature of our baseline transmitter experiments is that they seeded random variation in exposure to subtle, payoff-irrelevant reliability cues. One of our reliability manipulations—the *modular* manipulation—varied whether an otherwise-identical text contained certainty-denoting prefixes, uncertainty-denoting prefixes, or *no* reliability prefixes. Consider two respondents, A and B, who both heard a recording about housing followed by a low-reliability recording about retail, and consider their transmission of the retail recording. Suppose that respondent A heard a neutral-reliability housing recording, while B heard a high-reliability housing recording. Respondents A and B have both heard exactly the same recording about retail; but relative to respondent A, B has both heard more reliability references in total and has potentially benefited from a contrast effect between the high- and low-reliability recordings which could have increased the salience of reliability. We hence treat respondent B as being incidentally cued to think about reliability, while respondent A is not. Note that the cue here is *subtle* — consisting of reliability words seeded in a *different* recording than the one under consideration — and *payoff-irrelevant*, having nothing to do with the transmitter’s payoff for transmitting the retail recording.

Column (1) of Table 1 regresses an indicator for whether a transmitter’s message for a particular recording is unanimously classified by our LLM and human coders as transmitting reliability markers on an indicator for whether the *other* recording the transmitter heard contained reliability indicators. (The independent variable is 0 for transmitters whose other recording was a no-prefix *modular* recording, and 1 for all other transmitters.) The results show that hearing such payoff-irrelevant cues in the *other* recording increases the probability of transmitting reliability information for *this* recording by 10 percentage points, or nearly 50 percent. Columns (2)

and (3) confirm, using more granular LLM codings, that the effect is entirely driven by increased transmission of reliability indicators consistent with the original recording (high indicators for high-reliability recordings, and low indicators for low-reliability recordings).

Cues in economic TV news. To test the relevance of incidental reliability cues in a higher-stakes field setting, we examine data on economic cable news broadcasts. For the sake of brevity, we here briefly describe the main ideas behind our methodology and the main results, deferring a detailed exposition to Appendix E.

Our measure of the fidelity of uncertainty transmission on cable news is the correlation between the number of uncertainty-denoting words and phrases used in economic segments and a benchmark measure of “true” economic uncertainty calculated from newspapers (Baker et al., 2016). Our use of newspapers as a benchmark fits our previously-discussed distinction between written text—which is more premeditated and less subject to cognitive and memory constraints—and spontaneous word-of-mouth discussion or punditry, which is more vulnerable to the cognitive constraints we study. We test whether higher quantities of uncertainty language on a channel’s news segments *yesterday*—which we treat as cues that incidentally bring reliability to mind—are associated with a greater responsiveness of uncertainty language in economic segments to (our benchmark of) true economic uncertainty *today*. We find that this is in fact the case, and that this is robust to a range of specifications and placebo tests.

In other words, when pundits on a certain channel discussed uncertainty more yesterday, their discussion of uncertainty *today* is more sensitive to true uncertainty. This is consistent with the idea that recent experiences discussing uncertainty cue people to attend to it more, leading them to transmit it with higher fidelity. The fact that our outcome of interest is a slope means that serial correlation in uncertainty discussion would not by itself generate our results, and our use of day fixed effects (isolating between-channel variation in uncertainty discussion on preceding days) rules out effects of true economic uncertainty on attention to uncertainty. However, we emphasize that this analysis is only suggestive: for example, one alternative interpretation is that the act of communicating uncertainty on preceding days shifts commentators’ beliefs about the importance of transmitting uncertainty, rather than acting as an unconscious cue.

Cues in everyday conversations. We next turn to the British National Corpus, a dataset consisting of 1,251 recordings of everyday conversations between individuals in the UK, recorded at home or in other settings using participants’ smartphones (Love et al., 2017). We use a large language model to annotate each of the ~800,000 lines of usable conversational text in the corpus. We ask the model to identify lines containing questions; we further distinguish between questions concerning information that the speaker’s interlocutor would have direct access to (such as what happened to them that morning) and secondhand information (such as a weather forecast or piece of economic or political news). Such questions, by definition, contain a request for level information of some kind; we ask the model to additionally classify whether the question contains an explicit request for reliability information, such as the interlocutor’s certainty level or the source of the information. As we discuss below, such explicit requests are almost completely nonexistent. We therefore also code whether the question contains reliability markers or other

references to reliability that could act as indirect cues. Column (4) of Table 1 shows that a question containing such a marker is 21 percentage points more likely to be followed by an answer containing a reliability indicator. This remains true in Column (5), where we restrict to questions requesting second-hand information. Indirect cues of reliability in questions therefore raise the probability that the answer to the question mentions reliability.

This analysis is, again, only suggestive: it could also reflect omitted factors shifting reliability expressions in both questions and answers, or reliability cues shifting interlocutors’ beliefs about the importance of transmitting reliability information. However, the results of this analysis are consistent with the evidence from our experiments and TV news: even subtle cues appear to help trigger reliability transmission.

Table 1: Reliability Cues Increase Transmission of Reliability

	Experimental Results: Reliability Passed on (Handcoding)			British National Corpus: Answer Contains Reliability Indicator	
	(1) All Words (Unanimous)	(2) Aligned Words (LLM)	(3) Misaligned Words (LLM)	(4) All Questions	(5) Secondhand Info Questions
Other Recording Contained Reliability Prefixes	0.107*** (0.035)	0.100** (0.043)	-0.014 (0.030)		
Question Contained Reliability Prefixes				0.207*** (0.003)	0.224*** (0.003)
Constant	0.130*** (0.028)	0.274*** (0.037)	0.123*** (0.027)	0.151*** (0.002)	0.166*** (0.002)
Observations	702	702	702	90057	70751

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Note: Columns (1)-(3) report results from our baseline transmitter experiment. An observation is a transmitter-by-topic, restricted to transmitter-by-topics where the transmitter heard a high- or low-reliability recording about that topic. The independent variable in all three columns is an indicator for whether the recording the transmitter heard for the *other* topic was *not* a neutral-reliability modular recording. The dependent variable in Column (1) is an indicator for whether our LLM and human coders unanimously classify the transmitter’s transmitted message as containing reliability indicators. In Column (2), it is an indicator for whether the LLM classified the message as containing a reliability indicator aligned with the original message (high-reliability indicators for high-reliability messages, and vice versa; our human coders did not code at this level of granularity). In Column (3), it is an indicator for whether the LLM classified the message as containing a reliability indicator misaligned with the original message. Columns (4) and (5) use data from the British National Corpus. An observation is a line in one of the conversations in the corpus. Column (4) restricts to conversational lines that our LLM identifies as being questions, while Column (5) additionally restricts to questions that are about secondhand information the questioner’s interlocutor would not have direct access to (e.g., not questions about the interlocutor’s name or experience at a restaurant). The dependent variable in both columns is an indicator for whether the response to the question in the next conversational line contains any reliability indicators and the independent variable is an indicator for whether the question itself contains any reliability indicators.

Salient cues of reliability during transmission. As a final test of the importance of cues in determining the transmission of reliability information, we conduct an additional *high salience* experiment that increases the *during-transmission* salience of the distinction between the level and reliability of the original message. This experiment tests the hypothesis that differential information loss decreases when transmitters are directly reminded about the level–reliability distinction during the recording process, effectively equally cueing level and reliability. The design also allows us to assess whether emphasizing one dimension of information crowds out the transmission of the other.

The design closely follows the *explicit incentives* treatment described in Section 4.1.2, in which transmitters were explicitly incentivized to transmit both the level and reliability of the original message’s prediction. It adds three features to increase the salience of the level-reliability incentives at the time of recording: First, we add additional, more heavy-handed comprehension questions in which respondents need to correctly answer which types of information they need to transmit in the experiment. Second, just prior to each recording we ask respondents: “What do you have to pass on well to maximize your chances of receiving a bonus? Tick all that apply” with the following response options: (i) level of the speaker’s prediction; (ii) reliability of the speaker’s prediction. Respondents can only proceed once they correctly answer this question by selecting both. Third, on the actual recording page we add the following reminder: “Remember: Your bonus payment is based equally on how well you pass on both of the following: (i) The level of the speaker’s prediction. (ii) The reliability of the speaker’s prediction.” This reminder is presented in large, red font. The experiment was conducted on Prolific in November 2023 with 244 transmitters and 1,010 listeners.

Figure 4 summarizes the results. Consistent with the hypothesis that reliability information is more likely to come to mind when it is explicitly cued during transmission, Panel (a) shows that transmitters in this design talk much more about reliability, with nearly 80% of transmitted transcripts containing at least some information about the original prediction’s reliability, compared to just 30-40% in our previous experiments. The share of transcripts containing level information decreases, from 90-95% to 80-90%. Commensurately, Panels (b) and (c) document a large reduction in reliability information loss relative to our baseline experiment (from 90% to 40%), which is accompanied by a statistically significant increase in level information loss (from 36% to 53%). This pattern of results plausibly reflects crowding-out of level information as transmitters talk more about reliability, suggesting that the transmission of level and reliability information draw on common cognitive resources, such that increasing the transmission of one can come at the expense of the other.

Panel (c) shows that distortions of reliability information disappear entirely for weak-reliability messages but remain for strong-reliability messages. On the one hand, this may suggest that indicators of weak reliability are more salient or easier to transmit once transmitters have reliability in mind. On the other hand, this pattern may reflect a symmetric loss akin to the one documented before, coupled with an overall downward shift of perceived reliability that equally applies to all transmitted messages. Two pieces of evidence favor the latter explanation: our LLM and human handcoders unanimously find that reliability is equally likely to be transmitted in the high- versus low-reliability conditions, and a handcoding-based adjustment to the beliefs figure described in Appendix Section A.1.2 eliminates the apparent asymmetric loss.

Panel (d) documents the consequences for the overall pattern of listeners’ state belief updates. Transmission strongly attenuates belief movements towards zero on average. This is driven by the level information loss; moreover, the offsetting force of reliability information loss for weak-reliability messages, which pushed belief updates for those messages away from zero, is now absent (see the detailed discussion of forces in Section 3.3). As a result, transmission mostly

preserves the distinction between weak- and strong-reliability messages: listeners update less than half as much from weak-reliability compared to strong-reliability messages, regardless of whether they hear original or transmitted recordings. However, this also means that average belief updates from transmitted messages are shrunk even further than in our baseline experiment.

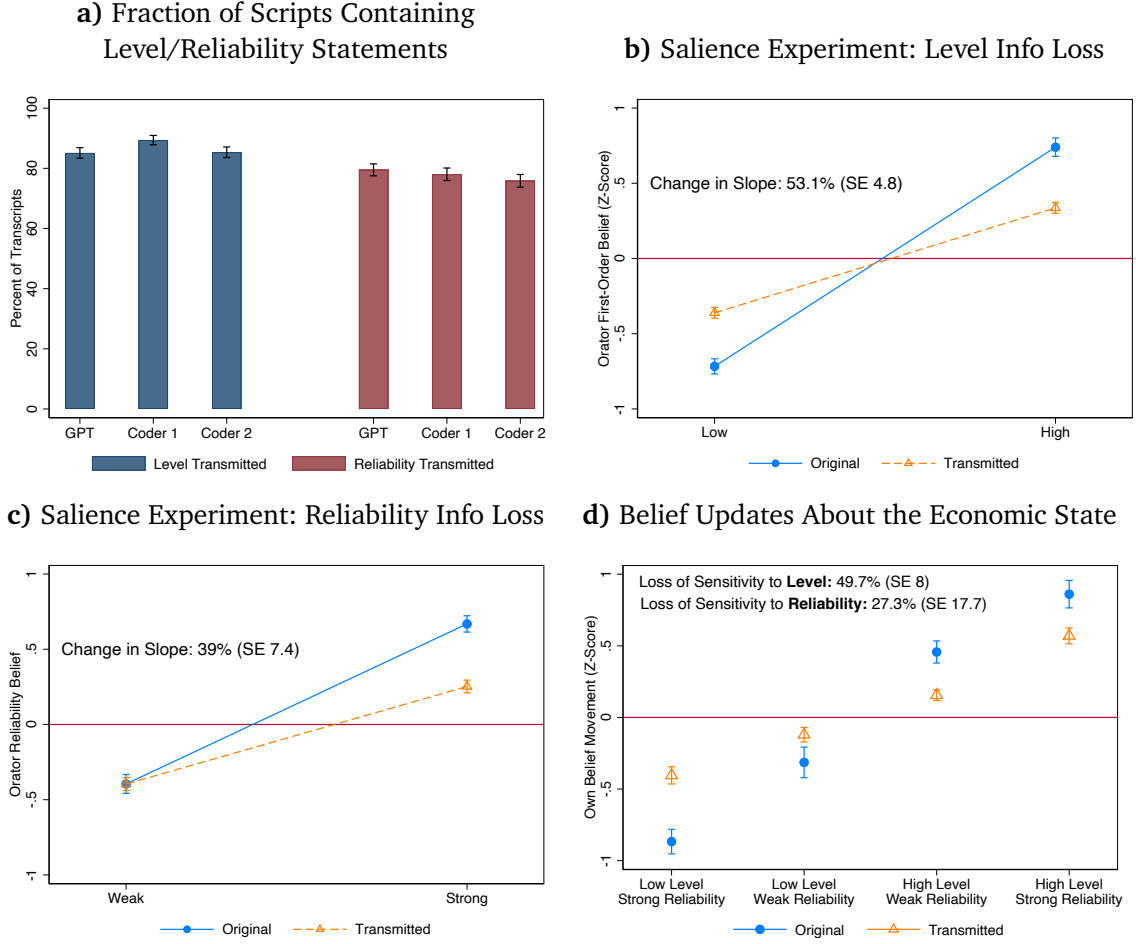


Figure 4: This figure presents data from the *high salience* experiment. Panel (a) replicates Panel (a) of Figure 1, showing which fraction of transmitted scripts contain statements about the level or reliability of the original prediction. Panels (b) and (c) replicate Figure 2, showing beliefs about the original message’s level and reliability, separately by whether level is low/high, reliability is weak/strong, and the listener is hearing an original or transmitted message. Panel (d) replicates Panel (c) of Figure 3, showing listeners’ average belief updates about the economic variable. Bars are standard error bars. $N = 1,010$ listeners and 244 transmitters.

Taken together, this experiment highlights a fundamental tradeoff. Making reliability salient at the moment of transmission restores the differential influence of weak- versus strong-reliability messages, but it does so by reallocating limited cognitive resources, thereby increasing level information loss and further attenuating overall belief updating. As a result, it aggravates the fact that in a population with heterogeneous priors, transmission loss slows down belief convergence on the basis of new information: belief updates in the intermediate quadrants of Figure 4 Panel (d) are more severely attenuated towards zero. Of course, such a slowdown may be desirable if

this convergence would otherwise happen on the basis of unreliable information.

Result 5. *Reminding transmitters to consider reliability at the moment of transmission substantially reduces reliability information loss, but does so by taxing a common cognitive resource, leading to increased loss of level information.*

Everyday requests for information rarely cue reliability. The preceding analysis suggests that cues are consequential. However, evidence from the British National Corpus suggests they are rare in practice. Of the 70,800 questions about secondhand information that the LLM identifies, only 2,911 (4%) contain an explicit request for information about the certainty or reliability of the interlocutor’s answer. Explicit requests for reliability information are hence very rare in everyday spoken conversation. We also ask the LLM to code whether each question contains any *subtler* cues of reliability, according to a maximally broad notion that we define as any language relating to certainty or possibility. Even according to this definition, less than half (46%) of questions contain such cues, compared to 19% of all conversational lines.

Cues: Summary. Across four pieces of evidence, cues that cause transmitters to attend to reliability—ranging from indirect, payoff-irrelevant cues to highly salient, payoff-relevant ones—robustly increase rates of reliability transmission. However, indirect cues of reliability are uncommon in spoken conversations, and explicit cues are very rare. This may lead generic transmission contexts to become more associated with level than reliability information, meaning that reliability tends not to be cued in such contexts, explaining failures to transmit it.

In Appendix C, we use our simulation exercises to explore the implications of cue-based reliability transmission for patterns of belief updating and polarization: cue-induced transmission of reliability consistently improves belief accuracy but can either increase or decrease polarization, depending on the nature of the cue and the axis along which polarization is measured.

4.3.3 Caveats and Discussion

This section has argued that the results of our experiments are best understood not as reflecting deliberate cost-benefit analyses supporting prioritization of level rather than reliability information, but involuntary cognitive mechanisms reflecting “what comes to mind.” Importantly, we do not claim that cost-benefit considerations are always absent in the field. To the contrary, such considerations may well amplify or attenuate the patterns we document, depending on context. Our claim is narrower: even abstracting from strategic motives, there is a systematic cognitive asymmetry that makes reliability less likely to be retrieved and verbalized during transmission. The experimental results establish this mechanism in isolation; the observational data show that the preconditions for this mechanism—infrequent cues of reliability—are pervasive in naturally occurring communication.

As previously mentioned, our finding that people neglect to transmit reliability information complements previous evidence that people neglect reliability when updating their beliefs (Griffin and Tversky, 1992). In the classical distinction between representational and computational stages of decision-making (see, e.g., Ba et al., 2024), current interpretations of reliability neglect

in belief updating focus on the *computational* stage: people attend to reliability information but either struggle to assess its quantitative magnitude (Augenblick et al., 2025) or imprecisely code and are uncertain how to incorporate it into belief-updating calculations (Enke et al., 2025). Our findings for the case of transmission suggest instead that reliability neglect here originates in the *representational* stage, with reliability failing to come to mind in the first place. This potential difference in mechanisms may not be surprising. Failures to bring reliability to mind when recalling a message from memory during transmission seem plausible, while such failures seem less likely to occur in controlled belief-updating problems where numeric measures of signal precision are explicitly presented to decision-makers. Conversely, computational difficulties seem more likely to arise in updating—when one must assess the quantitative magnitude of reliability and incorporate it into a Bayesian calculation—than in transmission—when one can simply repeat indicators of reliability.

One implication of the apparent origin of failures to transmit reliability in the representational stage of decision-making is that reliability transmission will be relatively insensitive to incentives or stakes, and more sensitive to contextual factors that shift a transmitter’s representation of the transmission problem at the moment of transmission. Even a payoff-irrelevant cue—one that has no claim on a cost-benefit calculation—substantially increases reliability transmission in our experiments, while explicit incentives to transmit reliability (in our Content Transmission Incentives experiment, where the incentives are introduced before the original recording is heard and several minutes before transmission) do not prevent the emergence of large differential loss. This pattern fits naturally with the Bordalo et al. (2025b) framework, in which incentives operate conditional on a representation: if reliability is absent from the default representation of the transmission problem, incentives delivered in advance of transmission may have limited purchase, while cues delivered at the moment of transmission can change the representation itself and hence be more impactful. A natural prediction, which our design does not directly test, and which we flag as a direction for future work, is that incentives delivered *at* the moment of transmission should behave more like contextual cues and hence be more effective.

5 Economic Consequences and Robustness

We close the paper by studying the economic consequences of the transmission loss we document and assessing the robustness of our experimental findings.

5.1 Consequences

5.1.1 Financial Investing Experiment

To assess whether transmission-induced losses of reliability information have economically meaningful consequences, we study a financial investment task. Decision making under risk, particularly investment, is a canonical economic setting where both level and reliability beliefs translate directly into consequential choices. Moreover, investment decisions are a context where the transmission of qualitative narratives has been argued to be particularly influential (Hirshleifer, 2020; Shiller, 2017), especially given the recent rise of social-media-driven stock and crypto investing.

This setting also lets us address a concern about interpretation. Our main outcomes so far are beliefs about the level and reliability of the original messages as well as incentivized state belief updates. While the latter are incentivized based on forecast accuracy, one might still worry that listeners’ reported assessments of reliability reflect a judgment of how reliable a message *sounds* rather than how willing they are to act on it. To address this concern, we use a setting in which listeners’ decisions translate directly into payoffs, so that perceived reliability is revealed through behavior with real stakes.

Design. This experiment departs from our baseline in three ways. First, the original narratives are about imminent earnings announcements for two real-world companies. We cross-randomize whether the original message argues that earnings will be higher or lower than the current consensus forecast. Second, transmitters are paid based on the realized performance of an incentivized investment decision made by the listener who hears their message. Specifically, listeners are tasked with allocating \$100 across assets that pay off depending on whether the relevant company’s earnings announcement ends up higher, lower, or within a small range around the current consensus forecast. One percent of listeners have their decisions implemented and the transmitters matched to those listeners receive the same payoffs as the listeners. Third, we vary the *uncertainty* (rather than reliability) expressed in the original messages by manipulating the speaker’s description of the distribution of possible outcomes. In the low-uncertainty case, the speaker says repeatedly that their prediction will certainly turn out to be true; in the high-uncertainty case, the speaker says repeatedly that the opposite could be true as well, and it’s possible to imagine things going either way. This allows us to test the robustness of our results to a new way of communicating reliability or uncertainty which is particularly natural in the investment setting.

Logistics. This preregistered experiment was conducted on Prolific in December 2025 with 299 transmitters and 1,287 listeners.

Results. The results, summarized in Panels (a) and (b) of Figure A13, and discussed in more detail in Appendix Section A.1.2, replicate our core finding: level information is lost at a rate of 14.5% and certainty information is lost at a rate of 77.4%. A formal test of equality of the two information loss statistics rejects the null at $p < 0.001$, $\chi^2 = 20.82$. Panel (c) shows that about 90% of transmitted messages contain level information, while only 50% contain reliability information. Panel (d) shows that listeners’ investment decisions react to information in a risk-averse way: they invest in the direction of the message they receive, shifting their investment more strongly in response to high-certainty messages than low-certainty messages. Transmission causes the sensitivity of the investment decision to level information to decrease by 20.6% and its sensitivity to certainty information to decrease by 59.3%, though these estimates are less precise due to the noisiness of the investment outcome.

Economic implications: back-of-the-envelope calculation. Using data on the beliefs and investment decisions of listeners who directly hear the original messages, we estimate listeners’ risk aversion assuming homogeneous CRRA preferences, estimating $\hat{\gamma} = 2.09$ (see Appendix A.1.2 for details; our estimate falls into the 1-3 range commonly found in the literature; see

Elminejad et al., 2025). We use our estimated utility function to calculate the welfare distortions implied by the level and certainty information loss we document. Eliminating both distortions would improve investment surplus by 19.5% in dollar terms and 6.5% in utility terms. Eliminating certainty information loss would improve surplus by 7.6% in dollar terms and 4.3% in utility terms (since the importance of reliability information is increasing in risk aversion).¹³

These magnitudes should be taken with a grain of salt, since they are sensitive to a range of required assumptions. Nevertheless, they provide a simple proof of concept that the information loss we document can matter nontrivially for welfare.

5.1.2 Diffusion of Information

The efficiency reductions documented above reflect a transmission-induced decline in the sensitivity of investment decisions to the certainty or reliability of information. Investment decisions could also be insufficiently sensitive to the reliability of new information due to private belief updating biases like those documented by Griffin and Tversky (1992) and Augenblick et al. (2025), among others. Appendix Section C describes the results of simple simulation exercises demonstrating that in the presence of such biases, transmission-induced loss of reliability information both compounds the insensitivity of decisions to the reliability of new information and—distinctively—partially shifts the distribution of efficiency losses onto sophisticated agents who *would* react accurately to reliability information had it reached them.

5.2 Robustness

We now summarize the design and results of three preregistered experiments probing the robustness of our main results to alternative design choices. Table 2 provides an overview of our results across all experiments. Full descriptions of experiments and results are available in Appendix B.

Extensive-margin transmission decisions. A valid concern is that our baseline results may partly be driven by the fact that transmitters *have to* pass on all messages. In the real world, individuals might simply choose not to pass on information they perceive as less credible, preventing unreliable information from diffusing and mistakenly being treated as reliable. To evaluate this possibility, we adapted the investment-transmission experiment described in Section 5.1 to allow transmitters the option of not recording a voice message.

Transmitters are given the same incentives as in the main investment design: their payoff is higher the greater the realized payoff of the investment made by their listener. As before, they are told that the listener will receive no information about the company other than their voice message. We explicitly tell respondents they have the option of not making a recording if they believe that the original message does not contain information that would be useful to pass on.

Appendix Figure A19 plots the share of transmitters choosing to record a message in each of the four level $\times$ certainty quadrants for the original message. Overall, transmitters choose to record a message 81% of the time, and this frequency barely varies across quadrants; when the

¹³By surplus, we mean the gains of eliminating information loss as a percentage of the gains of moving from the default no-information equal-allocation portfolio to the optimal portfolio.

original message is high-certainty, transmitters are more likely to record a message by a statistically insignificant 2.7 percentage points. When transmitters do choose to pass on a message, they mention reliability 42% of the time according to our GPT classification, somewhat lower than in the main investment experiment in Section 5.1.

The insensitivity of transmitters' sharing decisions to the certainty level of the original message reflects two factors. First, the certainty or reliability of the original information is not top-of-mind for transmitters deciding whether to record a message (see Appendix Table A7). This finding is consistent with evidence on the sharing of fake news on social media collected by Pennycook et al. (2021). Second, both in our experimental context and in the real world, choosing not to pass on a message is an ineffective remedy for the phenomenon we document. Declining to pass on a message avoids the potential downside of causing low-reliability information to be mistakenly interpreted as high-reliability (or vice versa), but it also means completely depriving the listener of any of the original information. This is reflected in the strong majority view among transmitters that passing on a message would be helpful to the listener.

Quantitative communication. Our baseline experiment used purely qualitative scripts because this imitates the majority of real-world communication. However, many important situations do involve the transmission of quantitative predictions or statements of numerical subjective probabilities. We therefore examine the robustness of our results to the addition of numerical statements about level and reliability to our original scripts. The experimental design is virtually identical to our baseline but adds quantitative information about both level and reliability to the original scripts. Quantitative information about the level is conveyed by providing a point estimate of the change in revenue growth. Quantitative reliability information is communicated via a probabilistic confidence statement. This experiment has the added benefit of alleviating potential concerns that our baseline results are driven by people perceiving our level manipulations as “more binary” or “more qualitative” than our reliability manipulations, and finding it easier to pass on binary or qualitative information. By communicating both level and reliability in exactly the same way at one point in the transcripts (through a single numerical percentage, e.g., an 8% increase in house price growth and a 90% confidence level), this experiment minimizes extraneous differences in the way level and reliability information are communicated.

Panels (a) and (b) of Appendix Figure A20 show that differential loss *strengthens* under this design: level information loss halves, to 12.8%, while reliability loss is unchanged. A formal test of equality of the two information loss statistics rejects the null at $p < 0.001$, $\chi^2 = 21.7$.

Transmission incentives based on the movement of belief distributions. Transmitters in our baseline experiment were incentivized to transmit messages that matched, as closely as possible, the mean belief updates induced by listening to the original messages. A potential concern is that this focus on *mean* belief updates might bring level information to mind more strongly than reliability information, whereas a broader focus on the *distribution* of beliefs would bring reliability to mind more easily.

To address this possibility, we replicate our main results with an alternative incentive scheme. Rather than eliciting listeners' mean beliefs, we elicit their full distribution of beliefs using a

graphical interface (Crosetto and De Haan, 2023). Transmitters are shown this interface, told how listeners will use it, and told that their probability of receiving the bonus payment depends on a measure of the distance between the average belief distribution induced by their message and the average belief distribution induced by the original message.

Panels (a) and (b) of Appendix Figure A21 show that differential information loss is unchanged under this design, at 33% for level and 88% for reliability, underscoring the robustness of differential loss of reliability information. A formal test of equality of the two information loss statistics rejects the null at $p < 0.01$, $\chi^2 = 6.91$.

Table 2: Comparison of Results across Incentive Schemes and Experimental Designs

Experiment	Incentive Scheme	% Scripts		Loss (SE)		<i>p</i>	Key Feature
		L	R	L	R		
<i>Panel A. Baseline</i>							
Belief Movement	Mean belief change	91	37	35.5 (4.4)	90.1 (4.5)	<.001	Core design
<i>Panel B. Alternative incentive schemes</i>							
Implicit Content	Holistic fidelity	90	35	31.0 (7.8)	73.7 (5.9)	<.001	Content transm. incentivized
Explicit Content	Level & reliab. ratings	90	35	35.6 (5.2)	65.3 (9.0)	<.001	Equal L&R incentive
Belief Distribution	Distribution shift	85	29	32.9 (18.3)	87.6 (7.5)	<.01	Beyond mean beliefs
Investment Payoffs	Shared listener payoff	90	50	14.5 (8.8)	77.4 [†] (10.3)	<.001	Aligned incentives
<i>Panel C. Design variations</i>							
Quantitative Scripts	Baseline scheme	91	53	14.2 (15.3)	88.7 (5.0)	<.001	Numerical statements
<i>Panel D. Mechanism experiment</i>							
High Salience	Explicit content + cue	85	78	53.1 (4.8)	39.0 (7.4)	<.05	Salient cues during transmission

Notes. L = level, R = reliability. “% Scripts” = share of transmitted messages classified (GPT) as containing level/reliability statements. Losses (%), SE in parentheses) from message beliefs via change-in-slope (Eq. 1); p tests equality of L and R losses. All designs are 2×2 in level (high/low) $\times$ reliability (strong/weak). [†]Certainty/uncertainty in the investment experiment. [‡]Among transmitters who chose to record.

6 Conclusion

Our economic decisions often rely on information sourced from others through verbal communication. Does the process of verbal transmission systematically distort economic information? We conduct a series of tightly controlled experiments to answer this question. Our experiments show that different types of information are subject to different degrees of transmission loss: the reliability of a prediction dissipates much more in the transmission process than the prediction’s level, reducing the influence of reliable information on beliefs, both in absolute terms and relative to unreliable information. Mechanism experiments demonstrate that reliability information is lost in transmission largely because it fails to come to mind during the transmission process, not because of gaps in perceived benefits or costs of transmitting level versus reliability information. A simple associative memory framework suggests that reliability is less likely to come to mind because it is less likely to be cued in everyday contexts. Examinations of everyday conversations

and economic TV news yield support for this possibility.

A striking feature of our evidence on cues is how little it takes to induce reliability transmission. Payoff-irrelevant prefixes in an unrelated recording, indirect linguistic cues in a conversation partner’s question, and even yesterday’s cable news coverage all shift reliability transmission today. This sensitivity to subtle, incidental contextual factors — combined with the relative insensitivity to incentives delivered in advance — is the signature of a representational mechanism in the sense of Bordalo et al. (2025b): reliability is absent from the default representation of the transmission problem but easily brought back in by context. The corresponding implication is that interventions to improve how reliable information diffuses need not strengthen incentives or comprehension; they need only ensure that reliability is part of how the transmission task is mentally represented at the moment of transmission.

Our findings speak to a variety of real-world phenomena, including viral fake news, persistent belief polarization, and failures of expert communication. The scope of our experimental results is, by necessity, limited. Longer delays between information consumption and transmission, multiple stages of transmission, and the richer set of incentives facing communicators in the real world may cause *even more extreme* differential information loss than what we document. Participants in our experiment are incentivized to faithfully convey the original information; by contrast, real-world communicators often face incentives to entertain, persuade, or impress, all of which may militate in favor of dropping caveats, admissions of uncertainty, or other reliability indicators. While incentives to persuade or entertain might result in even stronger reliability loss in real-world contexts, this may be counterbalanced by other factors, like the ability of listeners to ask follow-up questions probing the transmitter’s confidence.

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